

Software Project Proposal (SE 3112)

Project Title: আশ্বাস

Course Name: Software Project Lab
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This project proposal has been reviewed and approved.

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Contents

Project Overview	1
1.1 Introduction:	1
1.2 Problem Statement:	1
1.3 Objectives:.....	1
2 Proposed Solution :	1
2.1 Main Features	1
2.1.1 Structured Learning in Clinical Setting (The Course System)	1
2.1.2 Support & Monitoring	2
2.1.3 Digital Mental Health File	2
2.1.4 Expert Treatment	2
2.1.5 Community & Safety.....	3
2.1.6 Knowledge Hub	3
2.1.7 AI-Powered Personalization	3
2.1.8 Personalization Through AI	3
2.1.9 Accessibility & Engagement	3
2.2 Target users.....	3
2.3 Expected benefits	4
3.Development methodology.....	4
4. Technology stack	4
5. Work plan / timeline.....	4
6. Expected outcome.....	4
7.Limitations & Risk	4
8. Conclusion.....	5

Project Overview

1.1 Introduction:

আশ্বাস is an advanced mobile app that intends to change the world of mental health by offering structured digital therapy. This project seeks to create a safe and supportive environment that would allow individuals struggling with their emotions to benefit from the service without entering a clinical setting. The app targets several categories, including parents of special children, novice moms, single parents, professional employees, and students.

1.2 Problem Statement:

Today's mental health assistance has several main limitations like-

- No Continuous Monitoring: Therapy usually implies sessions that are not always followed up on.
- Hard to Entry: It is too expensive, and thus inaccessible to many people who fear the associated stigma.
- Need for a Platform: There is currently no way to connect junior therapists (interns/students) with senior doctors.

1.3 Objectives:

Objective 1: To develop a mood and mental well-being evaluation AI tool for real-time monitoring of users.

Objective 2: To incorporate a "Course & Assignment" system whereby users can submit their homework which will be assessed by experts.

Objective 3: To establish an anonymous forum and a support panel comprising interns and specialists available round the clock for real-time emotional counseling.

2 Proposed Solution :

আশ্বাস will function as a holistic system where the users go through a "Mental Health Path" rather than taking individual sessions.

2.1 Main Features

2.1.1 Structured Learning in Clinical Setting (The Course System)

Contrary to most apps that provide randomized content, আশ্বাস provides a structured system in the form of –

- Curriculum-Based Therapy: Users register for particular courses such as "Postpartum Wellness" or "Parenting for special needs".

- **Modular Learning:** A particular course includes a series of video tutorials, textual instructions, and required assignments.
- **Homework Portal:** A dedicated "Homework" portal wherein users share journal entries, audio notes, or images of their therapy sessions.
- **Progress Locking:** Completion of certain tasks or assignment to unlock subsequent modules.

2.1.2 Support & Monitoring

It is the underlying mechanism of the system, which guarantees that the user is never really alone.

- **24/7 Junior Support Queue:** A live panel staffed by psychology interns and junior doctors monitoring the user's behavior when senior specialists cannot handle requests.
- **Case Assignment:** The platform automatically assigns users to different levels of staff depending on the severity of their mood analysis.
- **Staff Feedback Process:** Staff evaluates the "Homework" submissions made by the user and gives personal feedback or comments on how to improve.
- **Monitoring Interface:** An interface for specialists to see the whole history of interaction between the patient and junior staff members.

2.1.3 Digital Mental Health File

- It works as an integrated digital record of the patient's health.
- **Visual Progress Report** – Graphical presentation of progress, number of completed courses and tasks.
- **Archive of Guidance** – Storage of all guidance and emotions shared by the user with the mental health care staff.
- **Emotional Analysis** – Long-term analytics of emotional changes of users. Can be downloaded and shared with a senior consultant

2.1.4 Expert Treatment

- **Telehealth Integration** – Easy booking for personal consultations with local and international therapists.
- **Consultant Selection** – Users' choice of a specialist according to preferred language (Bangla/English), his/her level of skill, specialization etc.
- **Automatic Notification** – Smart notification system reminding about booked appointments.

2.1.5 Community & Safety

Anonymous Support Hub: A private room for users where they can discuss their experiences and pose questions, while receiving "Empathy" emojis anonymously.

Quick Help: A button that allows the user to get access to a number of grounding practices or a way to contact an emergency hotline immediately and without navigating menus.

2.1.6 Knowledge Hub

A database of articles and videos for self-learning purposes, not tied to any particular training course.

2.1.7 AI-Powered Personalization

- Initial Psychological Test: An AI-based psychological assessment test recommending a particular Course Plan depending on a person's stress/anxiety/life situation.
- Mood Analysis: An interactive tool that allows users to select emoji and analyze their moods, resulting in a Mood Trend graph.
- Chat Assistant AI: The bot that responds to the user first and offers calm-down tips.

2.1.8 Personalization Through AI

- Smart Assessment – Initial AI-powered psychological assessment that recommends Course Plan according to one's stress, anxiety or category of their life.
- Daily Mood Monitoring – Interactive and emoji-driven mood slider that helps to draw information for the “Mood Trend” graph.
- AI Assistant Chat – First aid bot that answers urgent questions, gives soothing advice, and sums up one's condition to the human support.

2.1.9 Accessibility & Engagement

- Bilingual User Interface: The ability to change the interface language from English to Bangla (and vice versa).
- Dual Theme Support: The ability to switch between two types of theme: light mode (with pastel colors) and dark mode (purplish colors).
- Gamification Elements: Earning "Points" and "Badges" (such as "7-Day Streak", "Task Master") to motivate people during hard therapy days.

2.2 Target users

- 1 Parents with kids who have special needs.
- 2 First-time moms and single parents.
- 3 University students and corporate workers experiencing burnout.

2.3 Expected benefits

- Offers structured data-driven therapy to its users.
- Allows for reducing the costs of therapy due to the usage of junior support panels.
- Ensures the user feels "heard" 24/7 via interaction of AI and live employees.

3. Development methodology

Agile (Scrum) methodology: The development process will use an iterative development cycle to get continuous feedback from stakeholders and users.

Sprint 1: UI/UX design and requirements finalization.

Sprint 2: Core features development (Auth, Courses, and AI Chatbot).

Sprint 3: Integration of backend staff panel and testing.

4. Technology stack

Frontend: React Native

Backend: Node.js with Express.

Database: MySQL

AI implementation: Future Integration

5. Work plan / timeline

- Phase 1: Requirement analysis (Week 1-3): Feature lists definition/user research.
- Phase 2: Design (Week 4-6): Creating high-fidelity Figma prototypes of light and dark themes.
- Phase 3: Development (Weeks 7-10): Coding the front-end/backend and implementation of courses/assignments.
- Phase 4: Testing (Week 11): User acceptance testing/UAT and debugging based on feedback.
- Phase 5: Final submission (Week 12): Product deployment/final presentation of the project.

6. Expected outcome

A completed product will become a full-scale mobile application where users will be able to subscribe to psychological courses, monitor their mood, give therapeutic assignments, and receive recommendations from medical staff, forming their "Digital Mental Health File" in the end.

7. Limitations & Risk

- Time constraint: The development of an efficient course management software within seven weeks is a bit restrictive.

- Technical challenges: Making sure that the AI assistant maintains empathy and does not give dangerous suggestions.
- Security risk: Protection of confidential user psychological information through encryption.

8. Conclusion

আশ্বাস is not merely an application for meditation but a clinical management platform created especially for the modern world. Incorporating the teacher's idea of a "Staff-Patient-Course" connection makes this project unique as it allows giving support to people in need whatever their living condition.